



US012397778B2

(12) **United States Patent**
Kang et al.

(10) **Patent No.:** **US 12,397,778 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **APPARATUS AND METHOD FOR
AUTOMATIC PARKING BASED ON
PARKING AREA ENVIRONMENT
RECOGNITION**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 45 days.

(21) Appl. No.: **18/365,071**

(22) Filed: **Aug. 3, 2023**

(65) **Prior Publication Data**
US 2024/0123973 A1 Apr. 18, 2024

(30) **Foreign Application Priority Data**

Oct. 13, 2022 (KR) 10-2022-0131282
May 18, 2023 (KR) 10-2023-0064191

(51) **Int. Cl.**
G08G 1/14 (2006.01)
B60W 30/06 (2006.01)
B60W 30/18 (2012.01)
G06V 10/26 (2022.01)
G06V 20/58 (2022.01)

(52) **U.S. Cl.**
CPC **B60W 30/06** (2013.01); **B60W 30/18036**
(2013.01); **G06V 10/26** (2022.01); **G06V**
20/586 (2022.01); **B60W 2554/20** (2020.02)

(58) **Field of Classification Search**
CPC B60W 30/06; B60W 30/18036; B60W
2554/20; B60W 2720/24; G06V 10/26;
G06V 20/586
USPC 340/932.2, 933, 937, 988, 995.28
See application file for complete search history.

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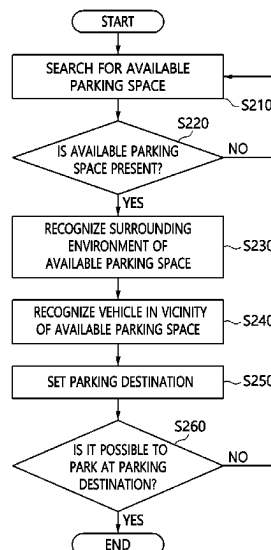
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(57) **ABSTRACT**

Disclosed herein are an apparatus and method for automatic parking based on recognition of a parking area environment. The method may include searching for an available parking space, determining whether reverse parking is possible by recognizing the environment of the available parking space, recognizing at least one additional vehicle located in the vicinity of the available parking space, and setting a parking destination based on the determination of whether reverse parking is possible and the result of recognition of the at least one additional vehicle.

15 Claims, 11 Drawing Sheets



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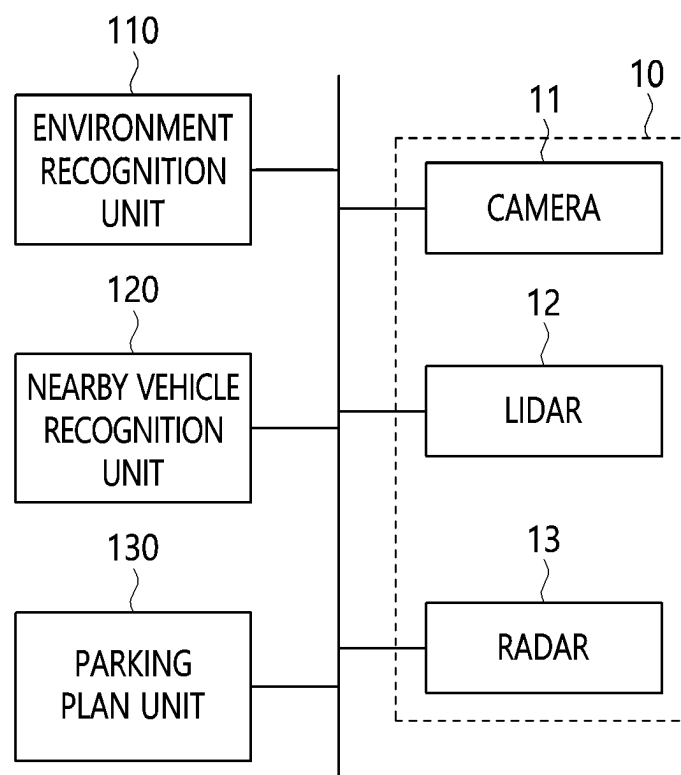
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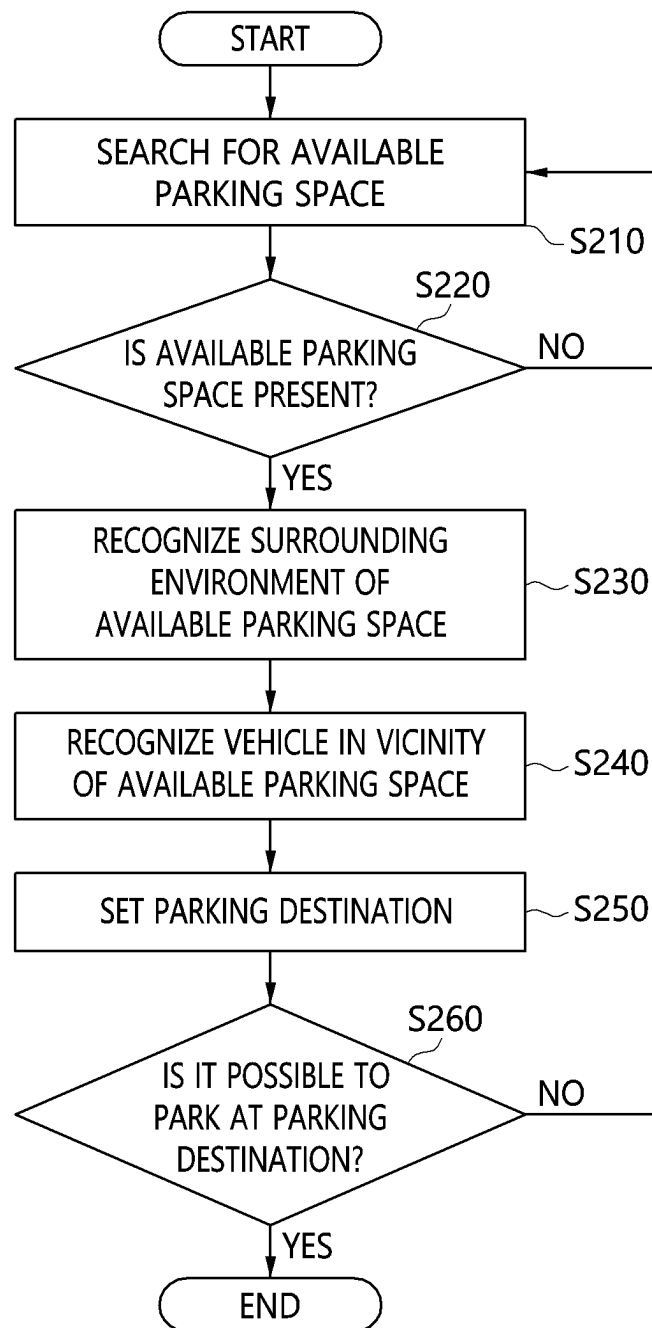
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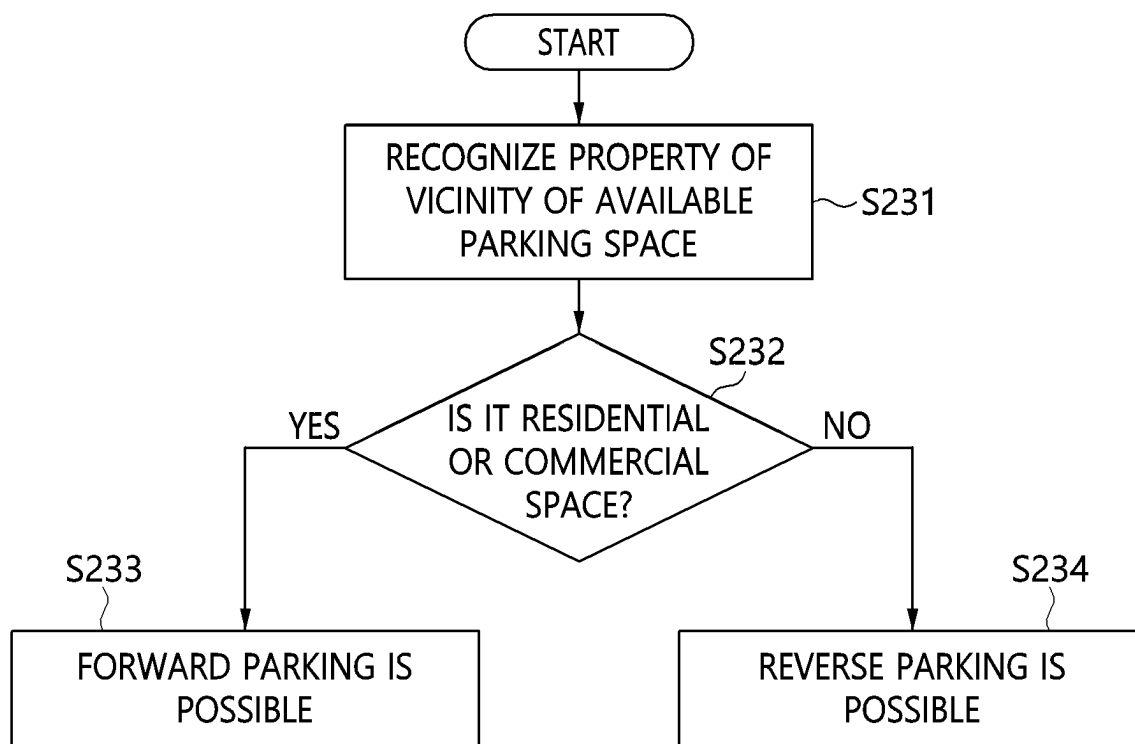
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100**FIG. 1**

**FIG. 2**

**FIG. 3**

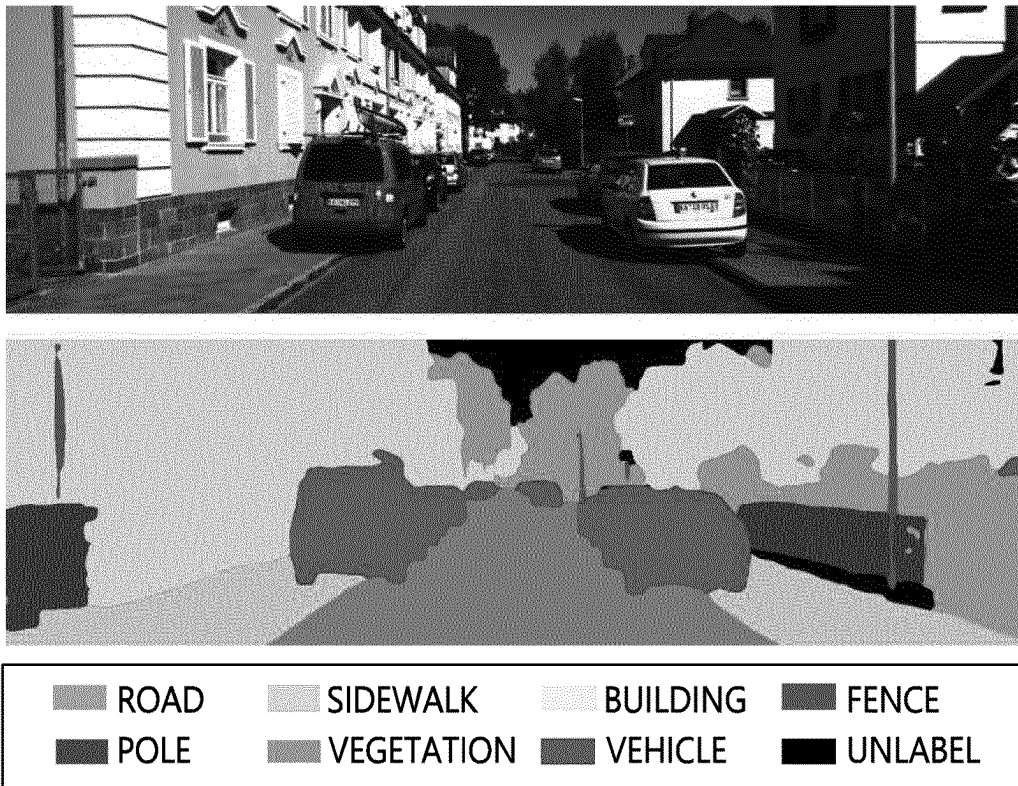


FIG. 4

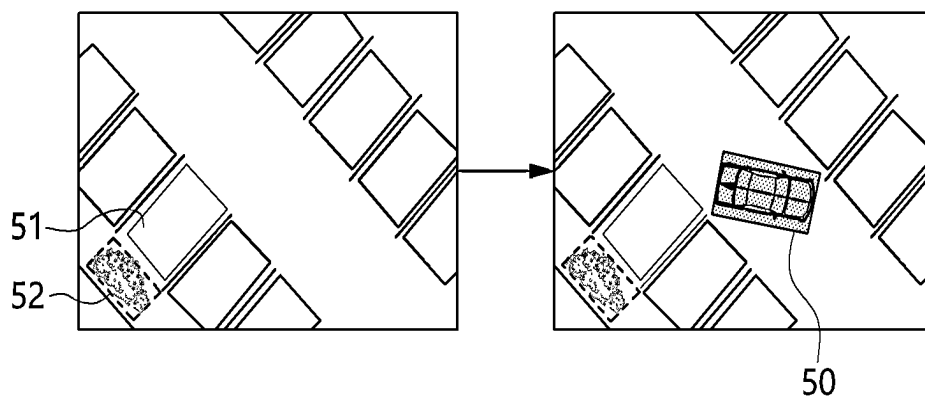


FIG. 5

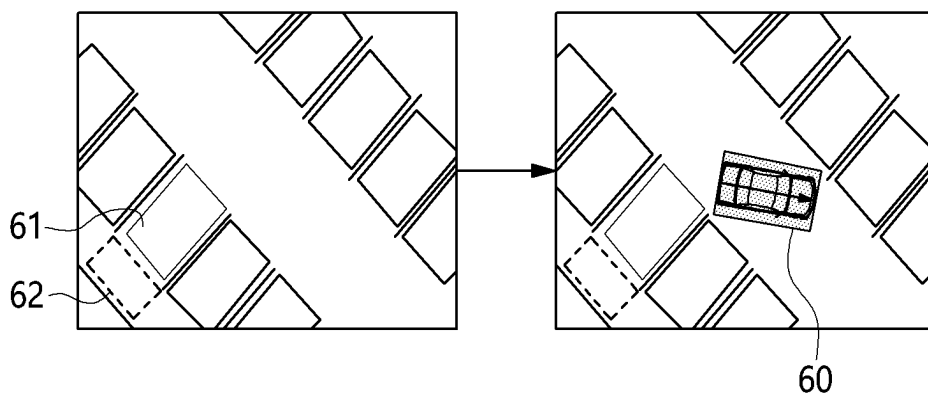
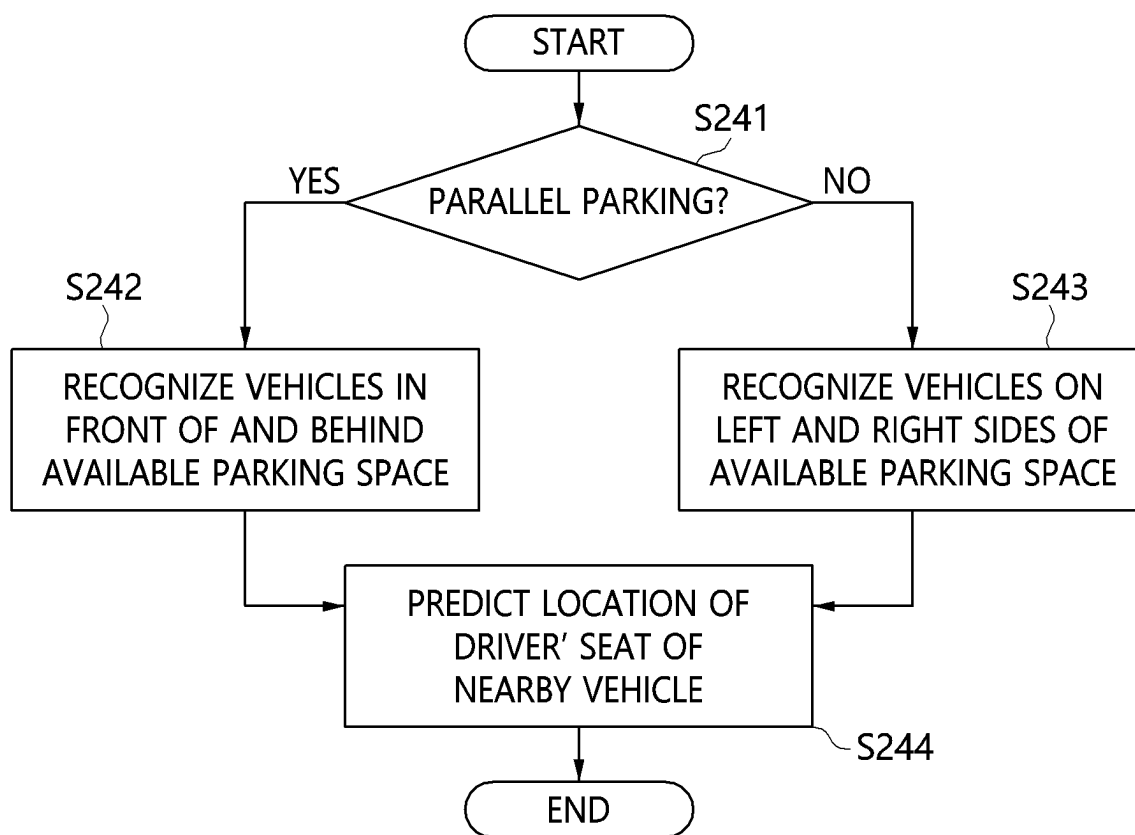


FIG. 6

**FIG. 7**

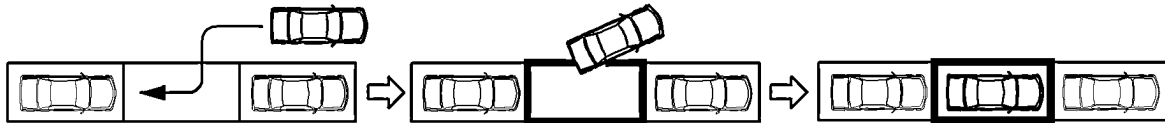


FIG. 8

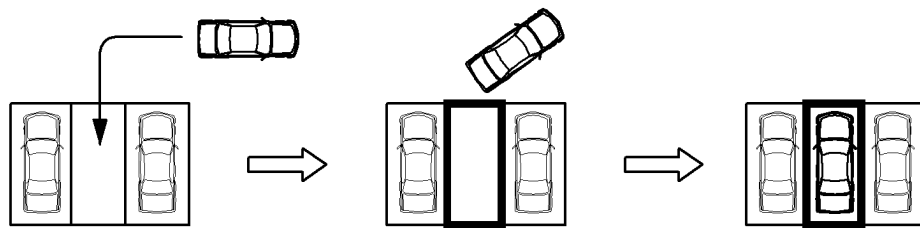


FIG. 9

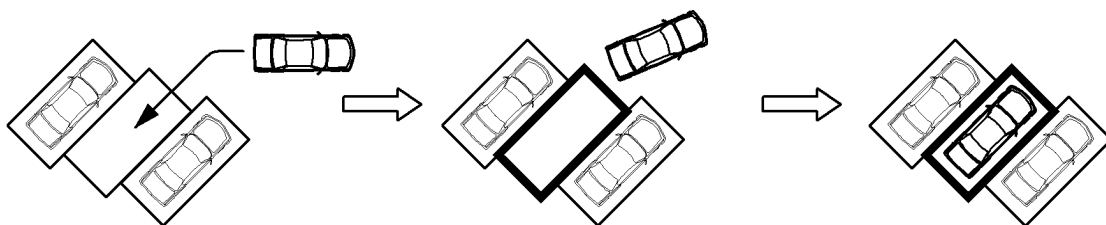


FIG. 10

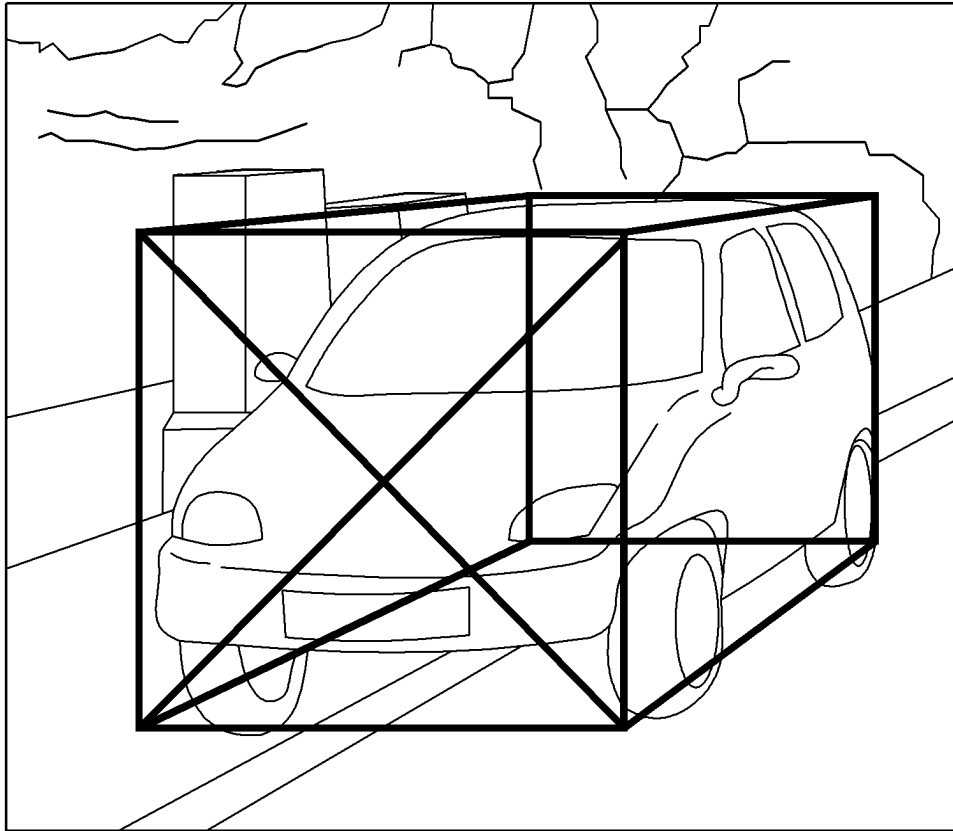


FIG. 11

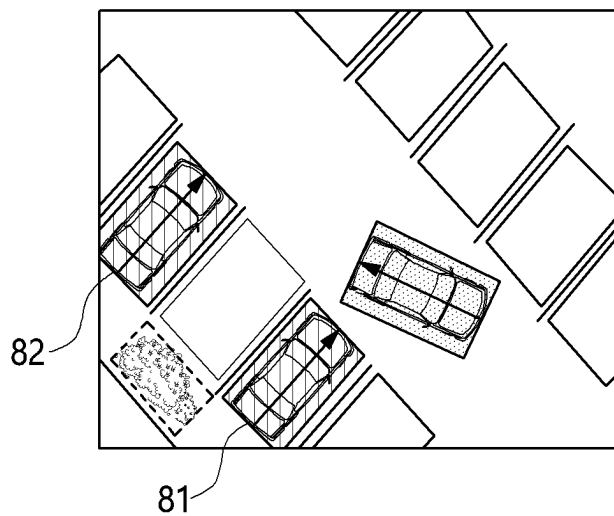


FIG. 12

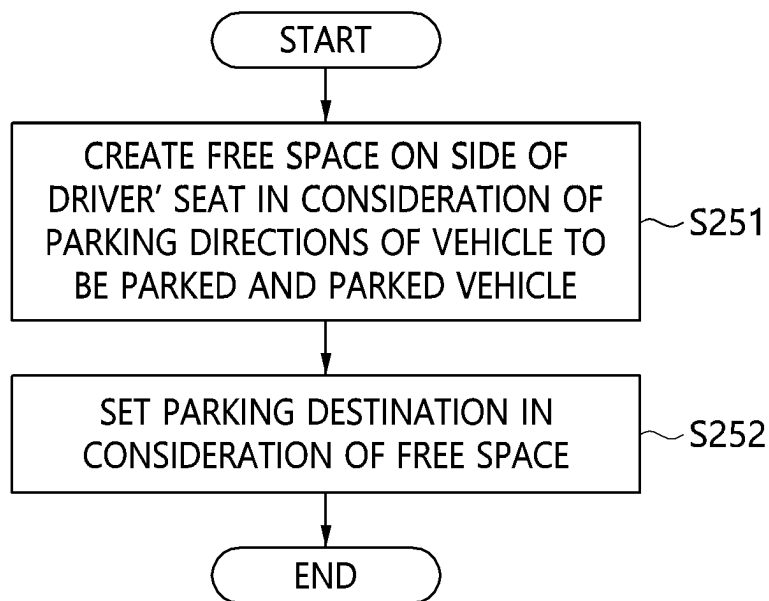


FIG. 13

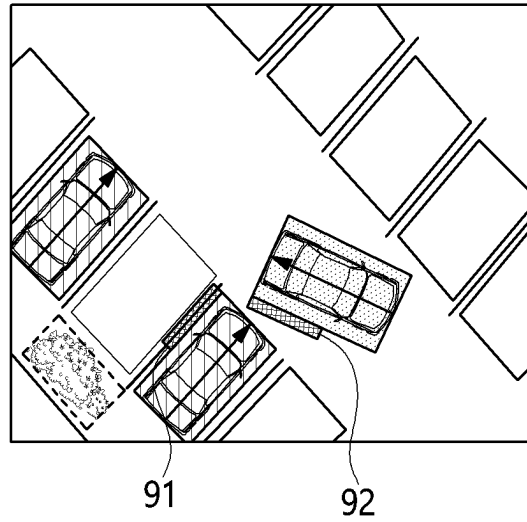


FIG. 14

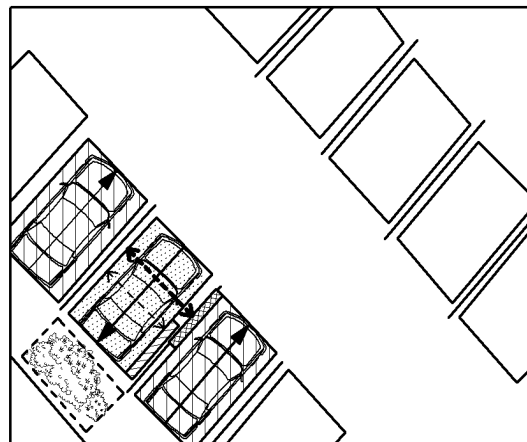


FIG. 15

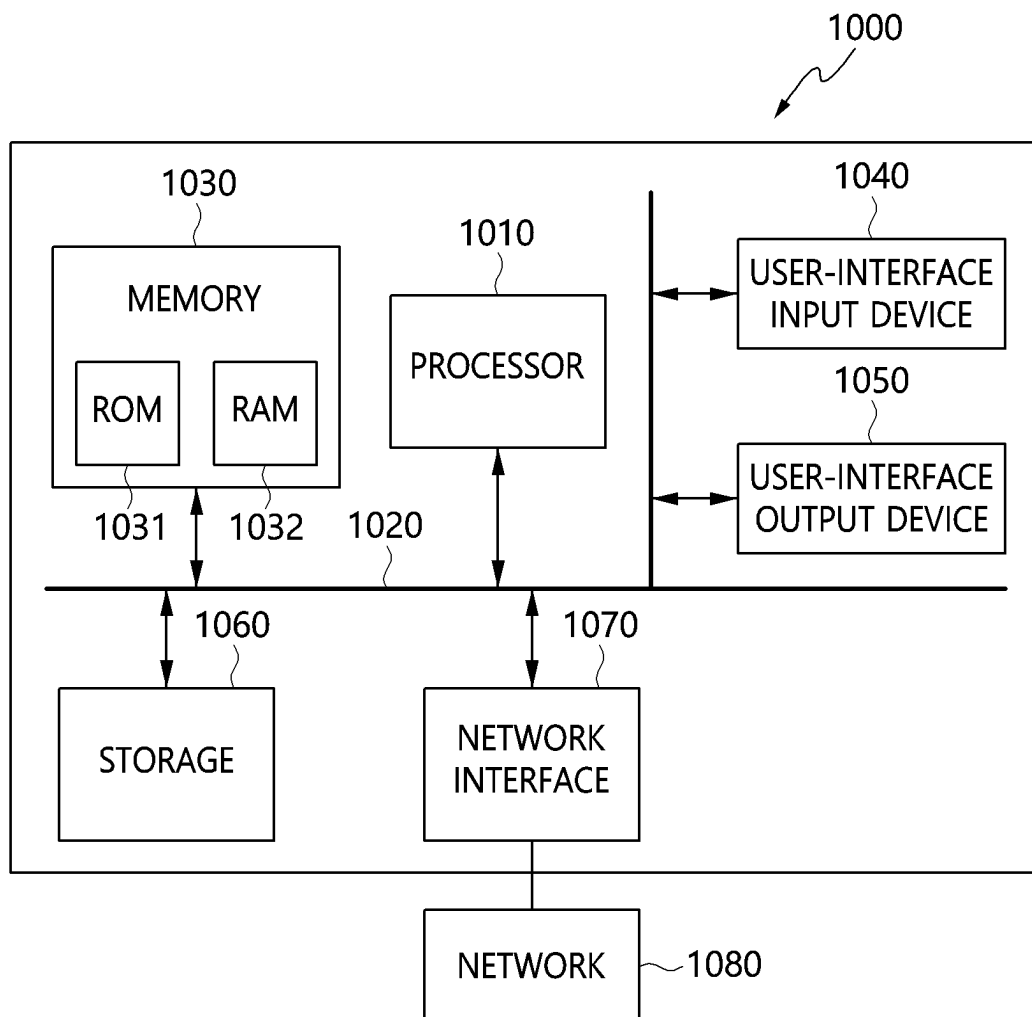


FIG. 16

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APPARATUS AND METHOD FOR AUTOMATIC PARKING BASED ON PARKING AREA ENVIRONMENT RECOGNITION

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of Korean Patent Applications No. 10-2022-0131282, filed Oct. 13, 2022, and No. 10-2023-0064191, filed May 18, 2023, which are hereby incorporated by reference in their entireties into this application.

BACKGROUND OF THE INVENTION

1. Technical Field

The disclosed embodiment relates to technology for automatic parking of an autonomous vehicle.

2. Description of the Related Art

Automatic parking technology is a system for searching for an available parking area using sensors installed in a vehicle and assisting with automatic parking and pulling out.

Parking assistance systems provide information about parking spaces, parking trajectories, and the like and functions such as steering control, and the like in the state in which a driver is inside a vehicle, and many vehicle manufacturers have commercialized such parking assistance systems. Automatic parking systems are technologies capable of parking by automatically controlling steering and braking without driver's intervention, and many companies are currently promoting commercialization thereof. Autonomous valet parking systems are technologies that enable vehicles themselves to search for available spaces in a parking lot and to park through autonomous driving in the state in which no driver is inside the vehicle, and development thereof is underway.

Meanwhile, there are various etiquette practices that should be observed when parking a vehicle. For example, when a target parking area is adjacent to a glass façade building or a garden, forward parking is widely recommended in order to avoid pollution from vehicle exhaust fumes.

Also, in order to prevent a door ding, which is a situation in which, when a driver or passenger gets in or out of a vehicle after parking, the door of the vehicle hits the side of an adjacent vehicle, and in order to enable the driver of the adjacent vehicle to more comfortably get in or out of his or her vehicle, it is recommended to leave more space on the side of the driver's seat of the adjacent vehicle rather than to park the vehicle in the center of the available parking space.

However, automatic parking technology of autonomous vehicles has primarily focused on safe parking so far, and the above-described parking etiquette practices that should be commonly observed have not been applied.

SUMMARY OF THE INVENTION

An object of the present disclosure is to observe common parking etiquette and improve the convenience of a driver getting in or out of a vehicle when automatic parking of an autonomous vehicle is performed.

A method for automatic parking based on recognition of a parking area environment according to an embodiment

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may include searching for an available parking space, determining whether reverse parking is possible by recognizing the environment of the available parking space, recognizing at least one additional vehicle located in the vicinity of the available parking space, and setting a parking destination based on the determination of whether reverse parking is possible and a result of recognition of the at least one additional vehicle.

Here, determining whether reverse parking is possible may include recognizing the property of the vicinity of the available parking space and determining that forward parking is possible when the property of the vicinity of the available parking space is a residential or commercial space.

Here, recognizing the property of the vicinity of the available parking space may comprise recognizing the property based on a semantic segmentation model that receives sensor data of a vehicle as input.

Here, recognizing the at least one additional vehicle may include recognizing the at least one additional vehicle located in the vicinity of the available parking space and recognizing the location and direction of the at least one additional vehicle.

Here, recognizing the at least one additional vehicle may comprise recognizing vehicles in front of and behind the available parking space in the case of parallel parking and recognizing vehicles parked on the left and right sides of the available parking space in the case of angle or perpendicular parking.

Here, recognizing the at least one additional vehicle may comprise recognizing the additional vehicle based on object detection technology using a three-dimensional (3D) bounding box.

Here, setting the parking destination may comprise predicting the locations of seats of drivers of a vehicle to be parked and the additional vehicle based on the directions of the vehicle to be parked and the additional vehicle and calculating the parking destination in consideration of free space on the sides of the predicted locations of the seats of the drivers.

An apparatus for automatic parking based on recognition of a parking area environment according to an embodiment includes memory in which at least one program is recorded and a processor for executing the program. The program may search for an available parking space, determine whether reverse parking is possible by recognizing the environment of the available parking space, recognize at least one additional vehicle located in the vicinity of the available parking space, and set a parking destination based on the determination of whether reverse parking is possible and a result of recognition of the at least one additional vehicle.

Here, when determining whether reverse parking is possible, the program may perform recognizing the property of the vicinity of the available parking space and determining that forward parking is possible when the property is a residential or commercial space.

Here, when recognizing the property of the vicinity of the available parking space, the program may recognize the property based on a semantic segmentation model that receives sensor data of a vehicle as input.

Here, when recognizing the at least one additional vehicle, the program may perform recognizing the at least one additional vehicle located in the vicinity of the available parking space and recognizing the location and direction of the at least one additional vehicle.

Here, when recognizing the at least one additional vehicle, the program may recognize vehicles in front of and

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behind the available parking space in the case of parallel parking and recognize vehicles parked on the left and right sides of the available parking space in the case of angle or perpendicular parking.

Here, when recognizing the at least one additional vehicle, the program may recognize the additional vehicle based on object detection technology using a 3D bounding box.

Here, when setting the parking destination, the program may predict the locations of seats of drivers of a vehicle to be parked and the additional vehicle based on the parking directions of the vehicle to be parked and the additional vehicle and calculate the parking destination in consideration of free space on the sides of the predicted locations of the seats of the drivers.

A method for automatic parking based on recognition of a parking area environment according to an embodiment may include searching for an available parking space, recognizing the property of the surrounding environment of the available parking space, determining whether reverse parking is possible depending on whether the property of the surrounding environment is a residential or commercial space, recognizing at least one additional vehicle located in the vicinity of the available parking space, recognizing the location and direction of the recognized at least one additional vehicle, predicting the locations of seats of drivers of a vehicle to be parked and the additional vehicle based on whether reverse parking is possible and the directions of the vehicle to be parked and the additional vehicle, and calculating a parking destination in consideration of free space on the sides of the predicted locations of the seats of the drivers.

Here, recognizing the property of the surrounding environment of the available parking space may comprise recognizing the property based on a semantic segmentation model that receives sensor data of a vehicle as input.

Here, recognizing the at least one additional vehicle may comprise recognizing the additional vehicle based on object detection technology using a 3D bounding box.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other objects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a schematic block diagram of an apparatus for automatic parking based on recognition of a parking area environment according to an embodiment;

FIG. 2 is a flowchart for explaining a method for automatic parking based on recognition of a parking area environment according to an embodiment;

FIG. 3 is a flowchart for explaining a step for recognizing a surrounding environment of an available parking space according to an embodiment;

FIG. 4 is an exemplary view of environment recognition based on semantic segmentation;

FIG. 5 is an exemplary view of forward parking according to an embodiment;

FIG. 6 is an exemplary view of reverse parking according to an embodiment;

FIG. 7 is a flowchart for explaining a step for recognizing other vehicles according to an embodiment;

FIG. 8 is an exemplary view of automatic parallel parking;

FIG. 9 is an exemplary view of automatic perpendicular parking;

FIG. 10 is an exemplary view of automatic angle parking;

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FIG. 11 is an exemplary view of object detection using a three-dimensional (3D) bounding box;

FIG. 12 is an exemplary view of prediction of the location and direction of an additional vehicle according to an embodiment;

FIG. 13 is a flowchart for explaining a step for recognizing other vehicles according to an embodiment;

FIG. 14 is an exemplary view of creation of free space according to an embodiment;

FIG. 15 is an exemplary view of setting of a parking destination in which free space is taken into account according to an embodiment; and

FIG. 16 is a view illustrating a computer system configuration according to an embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The advantages and features of the present disclosure and methods of achieving them will be apparent from the following exemplary embodiments to be described in more detail with reference to the accompanying drawings. However, it should be noted that the present disclosure is not limited to the following exemplary embodiments, and may be implemented in various forms. Accordingly, the exemplary embodiments are provided only to disclose the present disclosure and to let those skilled in the art know the category of the present disclosure, and the present disclosure is to be defined based only on the claims. The same reference numerals or the same reference designators denote the same elements throughout the specification.

It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements are not intended to be limited by these terms. These terms are only used to distinguish one element from another element. For example, a first element discussed below could be referred to as a second element without departing from the technical spirit of the present disclosure.

The terms used herein are for the purpose of describing particular embodiments only and are not intended to limit the present disclosure. As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Unless differently defined, all terms used herein, including technical or scientific terms, have the same meanings as terms generally understood by those skilled in the art to which the present disclosure pertains. Terms identical to those defined in generally used dictionaries should be interpreted as having meanings identical to contextual meanings of the related art, and are not to be interpreted as having ideal or excessively formal meanings unless they are definitively defined in the present specification.

FIG. 1 is a schematic block diagram of an apparatus for automatic parking based on recognition of a parking area environment according to an embodiment, and FIG. 2 is a flowchart for explaining a method for automatic parking based on recognition of a parking area environment according to an embodiment.

Referring to FIG. 1, the apparatus 100 for automatic parking based on recognition of a parking area environment

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according to an embodiment is installed in an autonomous vehicle, thereby performing operation of automatically parking and pulling out the vehicle.

Here, a sensor unit **10** for acquiring information about surroundings may be installed in the autonomous vehicle. The sensor unit **10** may include, for example, a camera **11**, a Light Detection and Ranging (LiDAR) device **12**, and radar **13**.

Here, the camera **11** and the LiDAR device **12** may acquire images of the surrounding environment of the vehicle, and the radar **13** may acquire the locations of objects in the surrounding environment of the vehicle and the distances to the objects.

The apparatus **100** for automatic parking based on recognition of a parking area environment acquires the surrounding environment information using the sensor unit **10** installed in the autonomous vehicle and performs automatic parking based on the acquired information.

Specifically, the apparatus **100** for automatic parking based on recognition of a parking area environment may include an environment recognition unit **110**, a nearby vehicle recognition unit **120**, and a parking plan unit **130**.

The environment recognition unit **110** recognizes the surrounding environment of the vehicle in order to perform autonomous driving or automatic parking. Particularly, parking requires more precise recognition than when general road driving is performed.

According to an embodiment, the environment recognition unit **110** searches for an available parking space and recognizes the environment of the found available parking space, thereby determining whether reverse parking is possible.

That is, referring to FIG. 2, the environment recognition unit **110** searches for a parking space based on sensor data sensed by the sensor unit **10** when the vehicle arrives at a destination at step **S210**.

When it is determined at step **S220** that an available parking space is present as the result of the search performed at step **S210**, the environment recognition unit **110** recognizes the surrounding environment of the available parking space at step **S230**. This will be described in detail later with reference to FIGS. 3 to 6.

Subsequently, the nearby vehicle recognition unit **120** recognizes at least one additional vehicle located in the vicinity of the available parking space at step **S240**. This will be described in detail later with reference to FIGS. 7 to 12.

Subsequently, the parking plan unit **130** sets a parking destination at step **S250** based on the determination of whether reverse parking is possible, which is output from the environment recognition unit **110**, and on the result of recognition of the at least one additional vehicle, which is output from the nearby vehicle recognition unit **120**. This will be described in detail later with reference to FIGS. 13 to 15.

Subsequently, when it is impossible to park at the set destination, the parking plan unit **130** goes to step **S210**.

FIG. 3 is a flowchart for explaining a step for recognizing a surrounding environment of an available parking space according to an embodiment, FIG. 4 is an exemplary view of environment recognition based on sematic segmentation, FIG. 5 is an exemplary view of forward parking according to an embodiment, and FIG. 6 is an exemplary view of reverse parking according to an embodiment.

Referring to FIG. 3, the environment recognition unit **110** recognizes the property of the vicinity of an available parking space at step **S231**.

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Here, the environment recognition unit **110** recognizes each of objects in the vicinity of the available parking space based on a sematic segmentation model that receives data acquired from the camera **11** or LiDAR device **12** installed in the vehicle as input, as illustrated in FIG. 4.

Here, sematic segmentation is technology for classifying the input of the sensor unit **10** into defined classes in units of pixels, and may be used to predict the property of a specific area.

The environment recognition unit **110** determines that forward parking is possible at step **S233** when the property of the vicinity of the available parking space, which is recognized at step **S231**, is a residential or commercial space at step **S232**. For example, when the vicinity of the available parking space **51** is a residential or commercial space including a flower bed **52** or glass material, as illustrated in FIG. 5, it may be determined that forward parking **50** is possible.

Conversely, when the property of the vicinity of the available parking space **61**, which is recognized at step **S231**, is not a residential or commercial space at step **S232**, the environment recognition unit **110** determines that reverse parking **60** is possible at step **S234**. For example, when the vicinity **62** of the available parking space is neither residential nor commercial space, it may be determined that reverse parking is possible, as illustrated in FIG. 6.

FIG. 7 is a flowchart for explaining a step for recognizing other vehicles according to an embodiment, FIG. 8 is an exemplary view of automatic parallel parking, FIG. 9 is an exemplary view of automatic perpendicular parking, FIG. 10 is an exemplary view of automatic angle parking, FIG. 11 is an exemplary view of object detection using a 3D bounding box, and FIG. 12 is an exemplary view of prediction of the locations and directions of other vehicles according to an embodiment.

Referring to FIG. 7, the nearby vehicle recognition unit **120** recognizes at least one additional vehicle located in the vicinity of an available parking space at steps **S241** to **S243**.

Here, when parallel parking is possible in the available parking space at step **S241**, as illustrated in FIG. 8, the nearby vehicle recognition unit **120** recognizes other vehicles located in front of and behind the available parking space at step **S242**.

Conversely, when it is determined at step **S241** that perpendicular parking is possible in the available parking space, as illustrated in FIG. 9 or that angle parking is possible in the available parking space, as illustrated in FIG. 10, the nearby vehicle recognition unit **120** recognizes other vehicles located on the left and right sides of the available parking space at step **S243**.

Here, the nearby vehicle recognition unit **120** recognizes other vehicles located in the vicinity of the available parking space based on object detection technology using a 3D bounding box, as illustrated in FIG. 11.

Here, object detection is technology for searching for a target object in the form of 3D box using the input sensor, and aims to discover both the location and direction of the object in the 3D space.

Subsequently, the nearby vehicle recognition unit **120** predicts the location and direction of the at least one additional vehicle at step **S244**. For example, the locations and parking directions of other vehicles **81** and **82** located on the left and right sides of the available parking space for perpendicular parking are recognized, as illustrated in FIG. 12.

FIG. 13 is a flowchart for explaining a step for recognizing other vehicles according to an embodiment, FIG. 14 is

an exemplary view of creation of free space according to an embodiment, and FIG. 15 is an exemplary view of setting of a parking destination in which free space is taken into account according to an embodiment.

Referring to FIG. 13, the parking plan unit 130 creates free space in consideration of the parking direction of the vehicle to be parked and the parking direction of at least one additional vehicle at step S251.

That is, based on the parking direction of the vehicle to be parked, which is determined at step S230, and the parking direction of the at least one additional vehicle parked in the vicinity thereof, the locations of the driver's seats in the respective vehicles are predicted. For example, the driver's seat may be on the left side in South Korea.

Subsequently, the parking plan unit 130 creates free space based on the predicted locations of the driver's seats. For example, free space 91 and free space 92 are virtually created on the left side of the driver's seat of the parked vehicle and on the left side of the driver's seat of the vehicle to be parked, as illustrated in FIG. 14.

Accordingly, the vehicle to be parked is induced to have more space on the side of the predicted location of the driver's seat such that the driver of the vehicle to be parked and the driver of the parked vehicle easily get in or out of their vehicles.

Subsequently, the parking plan unit 130 sets the final parking destination at step S252 in consideration of the virtually created free space. That is, the parking destination is calculated in the state in which the additional virtual free space is secured in the direction of the predicted location of the driver's seat, as illustrated in FIG. 15. Consequently, the vehicle to be parked is induced to be parked at the location at which the distance to the driver's seat of the adjacent vehicle is greater than when the vehicle is parked in the center of the parking space.

FIG. 16 is a view illustrating a computer system configuration according to an embodiment.

The apparatus according to an embodiment may be implemented in a computer system 1000 including a computer-readable recording medium.

The computer system 1000 may include one or more processors 1010, memory 1030, a user-interface input device 1040, a user-interface output device 1050, and storage 1060, which communicate with each other via a bus 1020. Also, the computer system 1000 may further include a network interface 1070 connected to a network 1080. The processor 1010 may be a central processing unit or a semiconductor device for executing a program or processing instructions stored in the memory 1030 or the storage 1060. The memory 1030 and the storage 1060 may be storage media including at least one of a volatile medium, a non-volatile medium, a detachable medium, a non-detachable medium, a communication medium, or an information delivery medium, or a combination thereof. For example, the memory 1030 may include ROM 1031 or RAM 1032.

The disclosed embodiment may observe common parking etiquette and improve the convenience of a driver getting in or out of a vehicle when automatic parking of an autonomous vehicle is performed.

Although embodiments of the present disclosure have been described with reference to the accompanying drawings, those skilled in the art will appreciate that the present disclosure may be practiced in other specific forms without changing the technical spirit or essential features of the present disclosure. Therefore, the embodiments described above are illustrative in all aspects and should not be understood as limiting the present disclosure.

What is claimed is:

1. A method for automatic parking based on recognition of a parking area environment, comprising:
 - searching for an available parking space;
 - determining whether reverse parking is possible by recognizing an environment of the available parking space;
 - recognizing at least one additional vehicle located in a vicinity of the available parking space; and
 - setting a parking destination based on a determination of whether reverse parking is possible and a result of recognition of the at least one additional vehicle, wherein
 - recognizing the at least one additional vehicle includes recognizing the additional vehicle located in the vicinity of the available parking space and recognizing a location and direction of the additional vehicle, and
 - setting the parking destination includes predicting a location of a driver's seat of the additional vehicle based on the recognized direction of the additional vehicle and calculating the parking destination in consideration of free space on a side of the predicted location of the driver's seat of the additional vehicle.
2. The method of claim 1, wherein determining whether the reverse parking is possible includes
 - recognizing a property of the vicinity of the available parking space; and
 - determining that forward parking is possible when the property is a residential or commercial space.
3. The method of claim 2, wherein recognizing the property of the vicinity of the available parking space comprises recognizing the property based on a semantic segmentation model that receives sensor data of a vehicle as input.
4. The method of claim 1, wherein recognizing the at least one additional vehicle comprises recognizing vehicles in front of and behind the available parking space in a case of parallel parking and recognizing vehicles parked on left and right sides of the available parking space in a case of angle or perpendicular parking.
5. The method of claim 1, wherein recognizing the at least one additional vehicle comprises recognizing the additional vehicle based on object detection technology using a three-dimensional (3D) bounding box.
6. The method of claim 1, wherein setting the parking destination further includes predicting a location of a driver's seat of a vehicle to be parked based on a direction of the vehicle to be parked, and
 - calculating the parking destination includes calculating the parking destination in consideration of free space on a side of the predicted location of the driver's seat of the vehicle to be parked.
7. An apparatus for automatic parking based on recognition of a parking area environment, comprising:
 - memory in which at least one program is recorded; and
 - a processor for executing the program,
 wherein the program searches for an available parking space, determines whether reverse parking is possible by recognizing an environment of the available parking space, recognizes at least one additional vehicle located in a vicinity of the available parking space, and sets a parking destination based on a determination of whether reverse parking is possible and a result of recognition of the at least one additional vehicle, wherein
 - when recognizing the at least one additional vehicle, the program performs recognizing the additional vehicle

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located in the vicinity of the available parking space and recognizing a location and direction of the additional vehicle, and

when setting the parking destination, the program predicts a location of a driver's seat of the additional vehicle based on the recognized direction of the additional vehicle and calculates the parking destination in consideration of free space on a side of the predicted location of the driver's seat of the additional vehicle.

8. The apparatus of claim 7, wherein, when determining whether reverse parking is possible, the program performs recognizing a property of the vicinity of the available parking space; and determining that forward parking is possible when the property is a residential or commercial space.

9. The apparatus of claim 8, wherein, when recognizing the property of the vicinity of the available parking space, the program recognizes the property based on a semantic segmentation model that receives sensor data of a vehicle as input.

10. The apparatus of claim 7, wherein, when recognizing the at least one additional vehicle, the program recognizes vehicles in front of and behind the available parking space in a case of parallel parking and recognizes vehicles parked on left and right sides of the available parking space in a case of angle or perpendicular parking.

11. The apparatus of claim 7, wherein, when recognizing the at least one additional vehicle, the program recognizes the additional vehicle based on object detection technology using a three-dimensional (3D) bounding box.

12. The apparatus of claim 7, wherein, when setting the parking destination, the program further predicts a location of a driver's seat of a vehicle to be parked based on parking directions of the vehicle to be parked, and

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when calculating the parking destination, the program calculates the parking destination in consideration of free space on a side of the predicted location of the driver's seat of the vehicle to be parked.

13. A method for automatic parking based on recognition of a parking area environment, comprising:

searching for an available parking space;

recognizing a property of a surrounding environment of the available parking space;

determining whether reverse parking is possible depending on whether the property of the surrounding environment is a residential or commercial space;

recognizing at least one additional vehicle located in a vicinity of the available parking space;

recognizing a location and direction of the recognized at least one additional vehicle;

predicting locations of a driver's seat of the additional vehicle based on the direction of the additional vehicle; and

calculating a parking destination in consideration of free space on a side of the predicted locations of the driver's seat of the additional vehicle.

14. The method of claim 13, wherein recognizing the property of the surrounding environment of the available parking space comprises recognizing the property based on a semantic segmentation model that receives sensor data of a vehicle as input.

15. The method of claim 13, wherein recognizing the at least one additional vehicle comprises recognizing the additional vehicle based on object detection technology using a three-dimensional (3D) bounding box.

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